

Habib University



Course Title: EE/CE-172/222 - Digital Logic and Design

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Assignment No. 01

Release Date: 19.01.2026

Due by: 25.01.2026

Total points: 100

Points obtained:

Student Name:	Student ID:	Section:
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Purpose:

The purpose of this assignment is to help you practice the number representation; its conversion; and the arithmetic operations in different base formats. In addition, this assignment will aid you to understand the process of representing a signed number in different formats; parity bits; representing complements of a number; and performing BCD arithmetic.

Instructions:

1. This assignment should be done individually.
2. All questions should be answered in **black ink only**. (Extra sheets can be used)
3. Scan your answer sheet and upload it on LMS before the due date.

Grading Criteria:

- You can also be asked to give a viva where you will be judged whether you understood the question yourself or not. If you are unable to correctly answer the question you have attempted right, you may lose your marks.
- Zero will be given if the assignment is found to be plagiarized.
- Untidy work will result in a reduction of your points.

Late submission penalty:

- 1-day late submission – 20% deduction of the maximum allowable marks
- 2-days late submission – 40% deduction of the maximum allowable marks
- No submission will be accepted after two days of the original deadline

CLO Assessment:

This assignment assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	Apply r-base (binary, octal, decimal, hexadecimal numbers systems) to digital systems and carry out arithmetic operations and conversions	✓
CLO 2	Apply principles of Boolean Algebra to represent and build equivalent realizations of digital logic (circuits)	
CLO 3	Design combinational logic circuits using logic gates	
CLO 4	Design sequential systems using finite state machine methodology	

P#	Questions	Pts																																	
1	Convert the following numbers with the indicated bases to decimal: a) $(1101.011)_2$ b) $(ABC.D3)_{15}$ c) $(123.01)_4$ d) $(62.73)_{20}$	2x4																																	
2	Convert the following numbers a) $(33.3)_{10}$ to binary b) $(3.21)_4$ to octal c) $(1.333)_8$ to hexadecimal d) $(0101.100)_2$ to base-9	2x4																																	
3	An 8-bit register stores 10100101. Determine what value is stored in decimal in the register if the bits represent: a) Unsigned number b) Signed number (2's complement system) c) Signed number (1's complement system) d) BCD	2x4																																	
4	You're part of the team of the biggest snacks competitor in the market. They intend to launch a never seen before snack to give to the world and dominate the other companies. Unfortunately, the secret formula of your snack was leaked, and now your only hope to have a competitive advantage is to set the product at the perfect price. In order to ensure your price is not leaked, you decide to encode it to communicate to your team. You are required to use all these characters and no other to send your message: <table border="1"><tr><td>%</td><td>\$</td><td>#</td><td>@</td><td>!</td></tr></table> a) Create a suitable number system and fill in the following table up to 11. State your assumptions clearly. <table border="1"><tr><td>Actual</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>...</td></tr><tr><td>Coded</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>...</td></tr></table> b) Represent the price 419_{10} in your coded notation so you can send it to your team c) You receive a counter price “%####” in your number system from your team, what does this represent in decimal?	%	\$	#	@	!	Actual	0	1	2	3	4	5	6	7	8	9	10	11	...	Coded													...	4+2+2
%	\$	#	@	!																															
Actual	0	1	2	3	4	5	6	7	8	9	10	11	...																						
Coded													...																						

5	You are trying to do your math homework and ask your friend from another planet for help. They send you this back to try and help you out. $(33 + 51) \times 70 = 6510$ If your friend has given you a correct equation, what number system does their planet use? (Hint: Find out the base of the system)	6																																																				
6	Convert the decimal number 689 into binary the following two ways: • Convert directly to binary • Convert first to hexadecimal and then from hexadecimal to binary. Which method is faster? Justify your answer with reasoning. (Requirement: You must show the working of both the parts and then give your reasoning based on time and method used. Marks will only be awarded if you can justify your answer).	4																																																				
7	You are creating a new currency that works using the base-18 number system (by extending the 16-base system) for your new country. Your friend decides to open up the first shop in your new country. They decide to sell fancy burgers for $(F2F)_{18}$ that cost $(ABC)_{18}$ to make. - Write down the first 20 numbers in the new system. - Your friend gets an order to make $(AA)_{18}$ burgers. They do not yet have a calculator. Help them determine how much they should charge the customer. - If they sell $(F)_{18}$ burgers, how much profit did they make in total? How much profit per burger?	1+4+5																																																				
8	Convert the last three alphabets of your first name and the first three alphabets of your last name to hexadecimal system, according to the given table. <table border="1"><tr><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td><td>H</td><td>I</td><td>J</td><td>K</td><td>L</td><td>M</td></tr><tr><td>40</td><td>41</td><td>42</td><td>43</td><td>44</td><td>45</td><td>46</td><td>47</td><td>48</td><td>49</td><td>4A</td><td>4B</td><td>4C</td></tr></table> <table border="1"><tr><td>N</td><td>O</td><td>P</td><td>Q</td><td>R</td><td>S</td><td>T</td><td>U</td><td>V</td><td>W</td><td>X</td><td>Y</td><td>Z</td></tr><tr><td>4D</td><td>4E</td><td>4F</td><td>50</td><td>51</td><td>52</td><td>53</td><td>54</td><td>55</td><td>56</td><td>57</td><td>58</td><td>59</td></tr></table> For example, your name is “BABAR MAZA,” you will convert “BAR” from BABAR and “MAZ” from MAZA as follows. (Use uppercase letters) “BAR” ➔ 414051 “MAZ” ➔ 4C4059 (Let the first number be X1 and the last name be X2) - Using 16’s Complement method, perform the following: $X1 - X2$ - Using 15’s Complement method, perform the following: $X2 - X1$ - Convert both X1 and X2 to radix 8 and find their sum	A	B	C	D	E	F	G	H	I	J	K	L	M	40	41	42	43	44	45	46	47	48	49	4A	4B	4C	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	4D	4E	4F	50	51	52	53	54	55	56	57	58	59	2x2+4
A	B	C	D	E	F	G	H	I	J	K	L	M																																										
40	41	42	43	44	45	46	47	48	49	4A	4B	4C																																										
N	O	P	Q	R	S	T	U	V	W	X	Y	Z																																										
4D	4E	4F	50	51	52	53	54	55	56	57	58	59																																										

9	Using the r's complement method, where r is the radix of the numbers mentioned, find the following differences: a) $(450643)_8 - (256752)_8$ b) $(1101.101)_2 - (1011.001)_2$	3x2																												
10	Convert the following decimal numbers to BCD to solve: a) $123 + 456$ b) $523 - 239$ c) $28 + 6.12$ d) $0.127 + 1.99$	2x4																												
11	Convert the following to signed (2's Complement) 6-bit binary numbers and perform the given operations. Also mention if there is an overflow or not. a) $(-6)_{10} + (+5)_{10}$ b) $(-1)_{10} - (+7)_{10}$ c) $(+11)_{10} + (+20)_{10}$ d) $(-23)_{10} + (-13)_{10}$	2x4																												
12	What is the smallest binary integer that can be expressed with a 16-bit computer using a signed number? An unsigned number? What are the equivalent decimal and hexadecimal numbers?	2x3																												
13	Complete the following table using 8-bit binary numbers: <table><tr><th>Decimal</th><th>Signed Magnitude</th><th>Signed 1's Complement</th><th>Signed 2's Complement</th></tr><tr><td>-10</td><td></td><td></td><td></td></tr><tr><td>-128</td><td></td><td></td><td></td></tr><tr><td>-32</td><td></td><td></td><td></td></tr><tr><td>+64</td><td></td><td></td><td></td></tr><tr><td>+127</td><td></td><td></td><td></td></tr><tr><td>+0</td><td></td><td></td><td></td></tr></table>	Decimal	Signed Magnitude	Signed 1's Complement	Signed 2's Complement	-10				-128				-32				+64				+127				+0				1x6
Decimal	Signed Magnitude	Signed 1's Complement	Signed 2's Complement																											
-10																														
-128																														
-32																														
+64																														
+127																														
+0																														
14	Find out the even parity bit of the following binary numbers: I. 01010111 II. 10110100	1x2																												
15	Convert any ONE of the following: a) BCD number 0110100101110011 to excess-3 b) Binary 0100100111110101 to Gray Code	4																												